

Algebraic Groups

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August 26, 2025

Abstract

References:

- Fourier-Mukai Transforms in Algebraic Geometry (Derived Categories for Algebraic Geometry)

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1 Basic Definitions

1.1 Group Schemes in General

Definition 1.1.1 (Group Schemes)

Let k be a field. Let X be a scheme of finite type over k . We say that X is a group scheme if X is group object in the category $\mathbf{Sch}_k$.

Definition 1.1.2 (Commutative Group Schemes)

Let k be a field. Let G be a scheme of finite type over k . We say that G is a commutative group scheme if G is an abelian group object in $\mathbf{Sch}_k$.

Definition 1.1.3 (Homomorphisms)

Let k be a field. Let G, H be group schemes over k . Let $\varphi : G \rightarrow H$ be a morphism of schemes over k . We say that φ is a homomorphism of group schemes if φ is also a group homomorphism.

1.2 Group Schemes as a Functor

Let k be a field. Let R be a finitely generated k -algebra. Let X be a scheme over k . Recall that the R -points of X is the set

$$X(R) = \mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{Spec}(R), X)$$

Proposition 1.2.1

Let k be a field. Let R be a finitely generated k -algebra. Let G be a group scheme over k . Then $G(R)$ is a group.

When k is algebraically closed, then the closed points of a group scheme over k is equal to $G(k)$ and we get a group structure on the closed points.

Lemma 1.2.2

Let k be a field. Let G be a group scheme over k . Then G is a commutative group scheme if and only if $G(R)$ is abelian for all $R \in \mathbf{FAlg}_k$.

Now recall that we constructed a fully faithful functor $\mathbf{Sch}_k \rightarrow \mathrm{Hom}_{\mathbf{Cat}}(\mathbf{FAlg}_k, \mathbf{Set})$ that sends a scheme X to its functor of points $X(-) : \mathbf{FAlg}_k \rightarrow \mathbf{Set}$. When G is group scheme, $G(R)$ is a group for all $R \in \mathbf{FAlg}_k$. By fully faithfulness, to give a group scheme is the same as giving a functor $F : \mathbf{FAlg}_k \rightarrow \mathbf{Grp}$ that lies in the essential image of the fully faithful functor.

Definition 1.2.3 (Schematically Dense)

Let k be a field. Let X be a scheme of finite type over k . Let S be a subset of $X(k)$. We say that S is schematically dense in X if the map $\mathcal{O}_X(X) \rightarrow \prod_{s \in S} k(s)$ defined by $f \mapsto (f(s))_{s \in S}$ is injective.

Proposition 1.2.4

Let k be a field. Let G be a group scheme over k . Then the following are equivalent.

- $G(k)$ is dense in G .
- $G(k)$ is schematically dense in G .
- G is reduced and $G(k)$ is dense in the subspace of closed points of G .

Proposition 1.2.5

Let k be a field. Let G be a group scheme over k . Then there is a one to one correspondence

$$\{H \leq G \mid H(k) \text{ is schematically dense in } H\} \xrightarrow{1:1} \{S \leq G(k) \mid S \text{ is a closed subgroup of } G(k)\}$$

Proposition 1.2.6

Let k be a field. Let G be a group scheme over k . Then there is a one to one correspondence

$$\{H \leq G \mid H \text{ is a variety}\} \xleftrightarrow{1:1} \{S \leq G(k) \mid S \text{ is a closed subgroup of } G(k)\}$$

Corollary 1.2.7

Let k be a separably closed field. Let G be a group scheme over k . Then the above correspondence restricts to a one to one correspondence

$$\{H \leq G \mid H \text{ is a variety}\} \xleftrightarrow{1:1} \{S \leq G(k) \mid S \text{ is a closed subgroup of } G(k)\}$$

1.3 Properties of Group Schemes

Lemma 1.3.1

Let k be a field. Let X be a group scheme over k . Then X is separated.

Proposition 1.3.2

Let k be a field. Let G be a group scheme over k . Then the following are equivalent.

- G is irreducible.
- G is connected.
- G is geometrically connected.
- For all $x \in X$, $N(\mathcal{O}_{X,x})$ is a prime ideal.

1.4 Group Subschemes

Definition 1.4.1 (Group Subschemes)

Let k be a field. Let G be a group scheme over k . Let H be subscheme of G . We say that H is a subgroup of G if H is a group scheme over k and the inclusion homomorphism $i : H \rightarrow G$ is a homomorphism.

Lemma 1.4.2

Let k be a field. Let G be a group scheme over k . Let H be a subscheme of G . Then H is a subgroup of G if and only if for all $R \in \mathbf{FGAlg}_k$, $H(R)$ is a subgroup of $G(R)$.

Proposition 1.4.3

Let k be a field. Let G be a group scheme over k . Let H be a subgroup of G . Then H is closed in G .

Lemma 1.4.4

Let k be a field. Let G be a group scheme over k . If G_{red} is geometrically reduced, then G_{red} is a subgroup of G .

1.5 The Identity Component

Lemma 1.5.1

Let k be a field. Let G be a group scheme over k . Then the following are true.

- G is irreducible if and only if G is connected.
- G is the disjoint union of its irreducible components.

Definition 1.5.2 (Identity Component)

Let k be a field. Let G be a group scheme over k . Define the identity component G° of G to be the unique irreducible component of G that contains 1_G .

Proposition 1.5.3

Let k be a field. Let G be a group scheme over k . Then the following are equivalent.

- G is smooth.
- G° is smooth.
- $\mathcal{O}_{G,1_G}$ is a regular local ring.
- $\dim(T_{1_G}G) = \dim(G)$.
- G is geometrically reduced.
- For all $R \in \mathbf{FAlg}_k$ and I an ideal of R such that $I^2 = 0$, we have that the map $G(R) \rightarrow G(R/I)$ is surjective.

2 Examples of Group Schemes

2.1 The Additive Group

Definition 2.1.1 (The Additive Group)

Let k be a field. Define the additive group to be the affine scheme

$$\mathbb{G}_a = \text{Spec}(k[x])$$

Lemma 2.1.2

Let k be a field. Then $\mathbb{G}_a$ is a group scheme.

Lemma 2.1.3

Let k be a field. Then the associated functor of $\mathbb{G}_a$ is given by the forgetful functor $\mathbf{FAlg}_k \rightarrow \mathbf{Grp}$ given by $R \mapsto (R, +)$.

2.2 The Multiplicative Group

Definition 2.2.1 (The Multiplicative Group)

Let k be a field. Define the multiplicative group to be the affine scheme

$$\mathbb{G}_m = \text{Spec}(k[x, x^{-1}])$$

Lemma 2.2.2

Let k be a field. Then $\mathbb{G}_m$ is a group scheme.

Lemma 2.2.3

Let k be a field. Then the associated functor of $\mathbb{G}_m$ is given by the forgetful functor $\mathbf{FAlg}_k \rightarrow \mathbf{Grp}$ given by $R \mapsto (R^\times, \cdot)$.

2.3 The Group Scheme Associated to the n th Roots of Unity

Definition 2.3.1 (The Group Scheme Associated to the n th Roots of Unity)

Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Define the group scheme associated to the n th Roots of Unity

$$\mu_n = \text{Spec}\left(\frac{k[x]}{x^n - 1}\right)$$

Lemma 2.3.2

Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Then μ_n is a group scheme.

Lemma 2.3.3

Let k be a field. Then the associated functor of μ_n is given by the functor $R \mapsto \{r \in R \mid r^n = 1\}$ for a finitely generated k -algebra R .

2.4 The General Linear Group

Definition 2.4.1 (General Linear Group)

Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Define the general linear group to be the affine scheme

$$GL_n(k) = \text{Spec}\left(\frac{k[x_{1,1}, \dots, x_{n,n}]}{\det(x_{i,j})x - 1}\right)$$

Lemma 2.4.2

Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Let $R \in \mathbf{FGAlg}_k$. Then we have

$$GL_n(k)(R) = GL_n(R)$$

Definition 2.4.3 (The Group of Automorphisms)

Let k be a field. Let V be a finite dimensional vector space over k . Define the GL_V to be the group scheme associated to the functor $\mathbf{FGAlg}_k \rightarrow \mathbf{Grp}$ defined by

$$R \mapsto \text{Aut}_R(V \otimes_k R)$$

Lemma 2.4.4

Let k be a field. Let V be a finite dimensional vector space over k . Let $\phi : V \rightarrow k^n$ be a vector space isomorphism. Then for each $R \in \mathbf{FGAlg}_k$, the map $GL(V \otimes_k R) \rightarrow GL_n(R)$ defined by $T \mapsto \phi \circ T \circ \phi^{-1}$ is a natural isomorphism from $GL(V)$ to $GL_n(k)$.

Corollary 2.4.5

Let k be a field. Let V be a finite dimensional vector space over k . Then $GL(V)$ is an affine group scheme.

2.5 The Group Scheme Associated to a Vector Space

Definition 2.5.1 (The Functor Associated to a Vector Space)

Let k be a field. Let V be a finite dimensional vector space. Define the functor

$$V_a : \mathbf{FGAlg}_k \rightarrow \mathbf{Grp}$$

as follows.

- A finitely generated algebra R over k is associated to $V \otimes_k R$.
- For $\varphi : R \rightarrow S$ a morphism of k -algebras, $V_a(\varphi)$ is the morphism $\text{id}_V \otimes_k \varphi$

Lemma 2.5.2

Let k be a field. Let V be a finite dimensional vector space over k . Then V_a is a group scheme.

Proposition 2.5.3

Let k be a field. Let V be a finite dimensional vector space over k . Then there is an isomorphism of group schemes

$$V_a \cong \mathbb{G}_a^{\dim(V)}$$

2.6 The Affine Group Scheme of a Group Ring

Let R be a commutative ring. Let G be an abelian group. Recall that $R[G]$ is the group algebra whose elements are R -linear combination of elements of G and multiplication is induced by the binary operation of G .

Definition 2.6.1 (Group Scheme Associated to Group Algebra)

Let k be a field. Let G be an abelian group. Define the affine group scheme associated to the group algebra of G to be $\text{Spec}(k[G])$.

Definition 2.6.2 (Functor Associated to Group Algebra)

Let k be a field. Let G be an abelian group. Define the functor

$$D(G) : \mathbf{FGAlg}_k \rightarrow \mathbf{Grp}$$

as follows.

- For each k -algebra R , define $D(G)(R) = \text{Hom}_{\mathbf{Grp}}(G, R^\times)$.
- For each k -algebra homomorphism, $\varphi : R \rightarrow S$ is sent to the group homomorphism $\text{Hom}_{\mathbf{Grp}}(G, R^\times) \rightarrow \text{Hom}_{\mathbf{Grp}}(G, S^\times)$ which is post composition with φ .

Lemma 2.6.3

Let k be a field. Let G be an abelian group. Then the functor $D(G)$ represents the affine group scheme $\text{Spec}(k[G])$.

Proposition 2.6.4

Let k be a field. Let G be a finitely generated abelian group. Suppose that $\varphi : G \rightarrow \bigoplus_{i=1}^r \mathbb{Z}/b_i\mathbb{Z} \oplus \mathbb{Z}^n$ is a group isomorphism. Then φ induces an isomorphism of group schemes

$$\text{Spec}(k[G]) \cong \prod_{i=1}^r \mu_{b_i} \times \mathbb{G}_m^n$$

3 Affine Group Schemes

3.1 Basic Properties

Theorem 3.1.1 (Cartier 1962)

Let k be a field of characteristic 0. Let G be an affine group scheme over k . Then G is smooth.

3.2 Relation to Hopf Algebra

Proposition 3.2.1

Let k be a field. Let A be a finitely generated algebra over k . Let $\Delta : A \rightarrow A \otimes_k A$ be a k -algebra homomorphism. Then $(\text{Spec}(A), \text{Spec}(\Delta))$ is an affine group scheme over k if and only if (A, Δ) is a Hopf algebra.

Proposition 3.2.2

Let k be a field. There is an equivalence of categories

$$\text{Spec} : \mathbf{HFGAlg}_k \xrightarrow{\cong} \mathbf{AffGrpSch}_k$$

with inverse given by Γ .

Proposition 3.2.3

Let k be a field. Let G be a group scheme over k . There is a one to one correspondence

$$\{H \subseteq G \mid H \text{ is a subgroup of } G\} \xleftrightarrow{1:1} \{I \subseteq \Gamma(G, \mathcal{O}_G) \mid I \text{ is a Hopf ideal of } \Gamma(G, \mathcal{O}_G)\}$$

given by the correspondence between closed subschemes and ideals of an affine scheme.

3.3 Structure Theorem for Group Schemes

Definition 3.3.1 (The Associated Affine Group Scheme)

Let k be a field. Let G be a group scheme. Define the associated affine group scheme to be

$$G^{\text{aff}} = \text{Spec}(\Gamma(G, \mathcal{O}_G))$$

Proposition 3.3.2

Let k be a field. Let G be a group scheme. Then the following are true.

- G^{aff} is an affine group scheme.
- There is a natural isomorphism

$$\text{Hom}_{\mathbf{GrpSch}_k}(G, H) \cong \text{Hom}_{\mathbf{GrpSch}_k}(G^{\text{aff}}, H)$$

given by factorizing through the natural map $\phi : G \rightarrow G^{\text{aff}}$ for any affine group scheme H over k .

- $\ker(\phi)$ is anti-affine.

Corollary 3.3.3

Let k be a field. Let G be a group scheme. Then G is an extension of affine algebraic group by an anti-affine algebraic group in a unique way.

4 The Isomorphism Theorems for Group Schemes

4.1 The Case of Affine Group Schemes

4.2 Extending to General Group Schemes

5 Representation Theory of Group Schemes

5.1 Group Actions

Definition 5.1.1 (Group Actions)

Let k be a field. Let G be a group scheme over k . Let X be a scheme of finite type over k . A group action of G on X is a morphism $\mu : G \times X \rightarrow X$ such that the following diagram commutes:

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{m \times \text{id}_X} & G \times X \\ \text{id}_G \times \mu \downarrow & & \downarrow \mu \\ G \times X & \xrightarrow{\mu} & X \end{array} \quad \begin{array}{ccc} * \times X & \xrightarrow{\text{incl.}} & G \times X \\ & \searrow \cong & \downarrow \mu \\ & & X \end{array}$$

Lemma 5.1.2

Let k be a field. Let G be a group scheme over k . Let X be a scheme of finite type over k . Let $\mu : G \times X \rightarrow X$ be a morphism. Then μ is a group action if and only if $\tilde{\mu} : \tilde{G} \times \tilde{X} \rightarrow \tilde{X}$ is a natural transformation such that for all $R \in \mathbf{FAlg}_k$, $\mu(R) : G(R) \times X(R) \rightarrow X(R)$ is a group action.

Lemma 5.1.3

Let k be a field. Let G be a group scheme over k . Let X be a scheme of finite type over k . Let $\mu : G \times X \rightarrow X$ be a group action. Then μ is faithfully flat.

Lemma 5.1.4

Let k be a field. Let G be a group scheme over k . Let X be a scheme of finite type over k . Let $\mu : G \times X \rightarrow X$ be a group action. Then the following are true.

- If G is smooth, then μ is smooth.
- If G is finite, then μ is finite.

5.2 Representations of Affine Group Schemes

Definition 5.2.1 (Representation of a Group Scheme)

Let k be a field. Let G be a group scheme over k . A representation of G is a vector space V over k together with a natural transformation $\rho : G \rightarrow GL(V)$.

We have seen that for any group scheme H , there is a natural isomorphism

$$\text{Hom}_{\mathbf{GrpSch}_k}(G, H) \cong \text{Hom}_{\mathbf{GrpSch}_k}(G^{\text{aff}}, H)$$

so it suffices to consider affine group schemes.

Lemma 5.2.2

Let k be a field. Let G be an affine group scheme over k . Let V be a finite dimensional vector space over k . If $\rho : G \rightarrow GL(V)$ is a natural transformation, then ρ is a homomorphism of group schemes.

Lemma 5.2.3

Let k be a field. Let G be an affine group scheme over k . Let V be a finite dimensional vector space over k . Then there is a one to one correspondence

$$\{\rho : G \rightarrow GL(V) \mid \rho \text{ is a representation}\} \xrightarrow{1:1} \{\phi : G \times V_a \rightarrow V_a \mid \phi \text{ is a group action}\}$$

Definition 5.2.4 (Comodule)

Proposition 5.2.5

Correspondence between comodules and representations.

Proposition 5.2.6

Let k be a field. Let G be an affine group scheme over k . Let V be a vector space over k . Let $\rho : G \rightarrow GL(V)$ be a representation. Then ρ is a filtered union of its finite dimensional subrepresentation.

Therefore it suffices to consider finite dimensional representations of affine group schemes.

5.3 The Regular Representation

Definition 5.3.1 (Faithful)

Let k be a field. Let G be a group scheme over k . Let V be a finite dimensional vector space over k . Let $\rho : G \rightarrow GL(V)$ be a representation. We say that ρ is faithful if $\rho(R)$ is injective for all $R \in \mathbf{FAlg}_k$.

Definition 5.3.2 (The Regular Representation)

5.4 Linear Group Schemes

Definition 5.4.1 (Linear Group Schemes)

Let k be a field. Let G be a group scheme over k . We say that G is linear if there exists a finite dimensional vector space V over k together with a faithful representation $\rho : G \rightarrow GL(V)$.

Proposition 5.4.2

Let k be a field. Let G be a group scheme over k . Then G is a linear group scheme if and only if G is an affine group scheme.

6 Character Theory

6.1 The Group of Characters

Definition 6.1.1 (Characters)

Let k be a field. Let G be an affine group scheme over k . A character of G is a homomorphism

$$\chi : G \rightarrow \mathbb{G}_m$$

Recall that $\text{Spec}(\mathbb{G}_m) = k[x, x^{-1}]$. Since homomorphisms $G \rightarrow \mathbb{G}_m$ are given by k -algebra homomorphisms $\mathcal{O}_{\mathbb{G}_m}(\mathbb{G}_m) \rightarrow \mathcal{O}_G(G)$, we see that giving a character is equivalent to a k -algebra homomorphism $k[x, x^{-1}] \rightarrow \mathcal{O}_G(G)$, which is equivalent to giving an invertible element $a \in \mathcal{O}_G(G)$ such that $\Delta(a) = a \otimes a$.

Definition 6.1.2 (Group-like Elements)

Let k be a field. Let G be an affine group scheme over k . Let $g \in G$. We say that g is a group-like element of G if

$$\Delta(g) = g \otimes g$$

Lemma 6.1.3

Let k be a field. Let G be an affine group scheme over k . Then there is a one to one correspondence

$$\{\chi : G \rightarrow \mathbb{G}_m \mid \chi \text{ is a character}\} \xleftrightarrow{1:1} \{a \in G \mid a \text{ is a group-like element}\}$$

Definition 6.1.4 (The Group of Characters)

Let k be a field. Let G be an affine group scheme over k . Define the group of characters of G to be the set

$$\chi(G) = \{\chi : G \rightarrow \mathbb{G}_m \mid \chi \text{ is a character of } G\}$$

together with the binary operation on $\chi(G)$ defined as follows. Let χ, χ' be characters. For each $R \in \mathbf{FAlg}_k$, $(\chi + \chi')(R) : G(R) \rightarrow R^\times$ is the group homomorphism defined by

$$(\chi + \chi')(R)(g) = \chi(R)(g) \cdot \chi'(R)(g)$$

Lemma 6.1.5

Let k be a field. Let G be an affine group scheme over k . Then $\chi(G)$ is an abelian group.

6.2 Eigenspaces

Definition 6.2.1 (Representations Acting through Characters)

Let k be a field. Let G be an affine group scheme over k . Let V be a vector space over k . Let $\rho : G \rightarrow GL(V)$ be a representation. Let $\chi \in \chi(G)$ be a character of G . We say that G acts on V through χ if ρ factors through χ as

$$\begin{array}{ccc} G & \xrightarrow{\rho} & GL(V) \\ & \searrow \chi & \nearrow \text{incl.} \\ & \mathbb{G}_m & \end{array}$$

Definition 6.2.2 (Eigenspaces)

Let k be a field. Let G be an affine group scheme over k . Let V be a vector space over k . Let $\rho : G \rightarrow GL(V)$ be a representation. Let $\chi \in \chi(G)$ be a character of G . Define the eigenspace V_χ of V with respect to χ to be the largest subrepresentation of V acting through χ .

Proposition 6.2.3

Let k be a field. Let G be an affine group scheme over k . Let V be a vector space over k . Let $\rho : G \rightarrow GL(V)$ be a representation. Let $\chi \in \chi(G)$ be a character of G . Let $\rho^\vee$ be the corresponding coaction of ρ . Then we have

$$V_\chi = \{v \in V \mid \rho^\vee(v) = v \otimes a(\chi)\}$$

where $a(\chi)$ is the corresponding group-like element of χ .

Proposition 6.2.4

Let k be a field. Let G be an affine group scheme over k . Let V be a vector space over k . Let $\rho : G \rightarrow GL(V)$ be a representation. Let $\chi \in \chi(G)$ be a character of G . If $V = \sum_{\chi \in \chi(G)} V_\chi$ is a sum of eigenspaces, then

$$V = \bigoplus_{\chi \in \chi(G)} V_\chi$$

is a direct sum of eigenspaces.

6.3 Diagonalizable Groups

Definition 6.3.1 (Diagonalizable Groups)

Let k be a field. Let G be an affine group scheme over k . We say that G is diagonalizable if

$$\mathcal{O}_G(G) = k\langle g \in G \mid g \text{ is a group-like element} \rangle$$

as k -vector spaces.

Definition 6.3.2 (The Category of Diagonalizable Group Schemes)

Define the category **Diag** of diagonalizable affine group schemes over k to be the full subcategory of **AffGrpSch** that consists of diagonalizable affine group schemes.

Lemma 6.3.3

Let k be a field. Let G be an affine group scheme over k . Then G is diagonalizable if and only if there exists a finitely generated abelian group A such that $G \cong \text{Spec}(D(A))$ as group schemes.

Definition 6.3.4

Let k be a field. Define the functor

$$D : \mathbf{FGAb} \rightarrow \mathbf{Diag}_k$$

by the following data.

- For each finitely generated abelian group G , $D(G)$ is the group scheme $\text{Spec}(k[G])$.
- For each group homomorphism $\phi : G \rightarrow H$, $D(\phi)$ is homomorphism of group schemes induced by the natural map $k[G] \rightarrow k[H]$ by the universal property of the k -algebra $k[G]$.

Proposition 6.3.5

Let k be a field. Then the following are true.

- $D : \mathbf{FGAb} \rightarrow \mathbf{Diag}_k$ is an equivalence of categories, with inverse given by χ sending a group scheme to its abelian group of characters.
- D is an exact functor.

Corollary 6.3.6

Let k be a field. Let G be a diagonalizable affine group scheme over k . If $H \leq G$ is a subgroup of G , then H and G/H are diagonalizable.

6.4 Diagonalizable Representations

Definition 6.4.1 (Diagonalizable Representations)

Let k be a field. Let G be an affine group scheme over k . Let $\rho : G \rightarrow GL(V)$ be a representation of G . We say that ρ is diagonalizable if ρ is the sum of 1-dimensional representations.

Proposition 6.4.2

Let k be a field. Let G be an affine group scheme over k . Then the following are equivalent.

- G is diagonalizable.
- Every finite dimensional representation of G is diagonalizable.
- Every representation of G is diagonalizable.
- For every representation $\rho : G \rightarrow GL(V)$ of G , there is a decomposition

$$V = \bigoplus_{\chi \in \chi(G)} V_{\chi}$$

7 Group Varieties

7.1 Algebraic Groups

Definition 7.1.1 (Algebraic Groups)

Let k be field. Let V be a geometrically reduced variety over k . We say that V is an algebraic group (or a group variety) if V is also a group scheme.

Definition 7.1.2 (Algebraic Subgroups)

Let k be a field. Let G be an algebraic group. Let $H \subseteq G$ be a subset of G . We say that H is an algebraic subgroup of G if H is an algebraic group and the inclusion map $i : H \rightarrow G$ is a homomorphism.

Lemma 7.1.3

Let G be an algebraic group. Then G is smooth.

Proof Since the set of all smooth points of a variety is an open subset, G must have at least one smooth point. Moreover, the translation morphism $l_y : G \rightarrow G$ defined by $x \mapsto yx$ is an isomorphism of varieties. Hence every point of G is smooth. ■

Lemma 7.1.4

Let G be an algebraic group. Let $H \leq G$ be a subgroup of G . Then the following are true.

- If H is closed, then H is an algebraic group.
- $\overline{H}$ is an algebraic group.

Lemma 7.1.5

Let G, H be algebraic groups. Then $G \times H$ is an algebraic group.

Lemma 7.1.6

Let G, H be algebraic groups. Let $\varphi : G \rightarrow H$ be a homomorphism. Then the following are true.

- $\ker(\varphi)$ is a closed subgroup of G .
- $\text{im}(\varphi)$ is a closed subgroup of H .
- $\dim(G) = \dim(\ker(\varphi)) + \dim(\text{im}(\varphi))$.

7.2 The Identity Component

Lemma 7.2.1

Let G be an algebraic group. Then 1_G is contained in exactly one irreducible component of G .

Definition 7.2.2 (The Identity Component)

Let G be an algebraic group. Define G° to be the unique irreducible component that contains 1_G .

Proposition 7.2.3

Let G be an algebraic group. Then the following are true.

- G° is a normal subgroup of G .
- $[G : G^\circ] < \infty$.
- G/G° is precisely the set of connected irreducible components of G .
- Let $H \leq G$ be a subgroup such that $[G : H] < \infty$. Then $G^\circ \subseteq H$.

Lemma 7.2.4

Let G, H be algebraic groups. Let $\varphi : G \rightarrow H$ be a homomorphism. Then we have

$$\varphi(G^\circ) = \varphi(G)^\circ$$

Definition 7.2.5 (Connected Algebraic Groups)

Let G be an algebraic group. We say that G is connected if $G = G^\circ$.

In other words, we say that G is connected if G is irreducible.